	práctico 6 - Campo magnético	HOJA N°:
	Física II	FECHA:

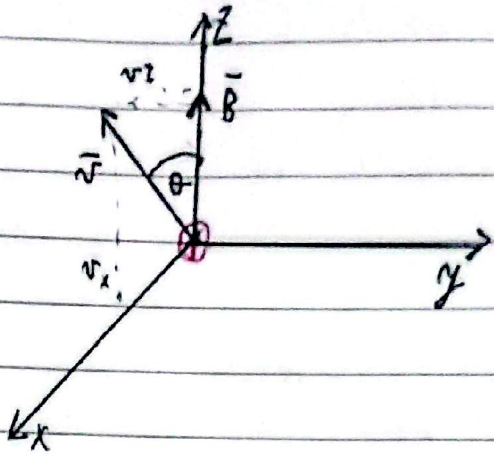
1)-

$q = 1,6 \cdot 10^{-19} C$

$\vec{v} = 3 \cdot 10^{5}\ m/s$

$\vec{B} = 2T$

$\theta = 30°$



$\vec{F_B} = q \vec{v} \times \vec{B}$

$v_x = v \cdot Sen(30°) = 150000\ m/s$

$v_z = v \cdot cos(30°) = 259807\ m/s$

$$\bar{v} \times \bar{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 150000 & 0 & 259807 \\ 0 & 0 & 2 \end{vmatrix} = -3 \cdot 10^{5}\ \hat{j}$$

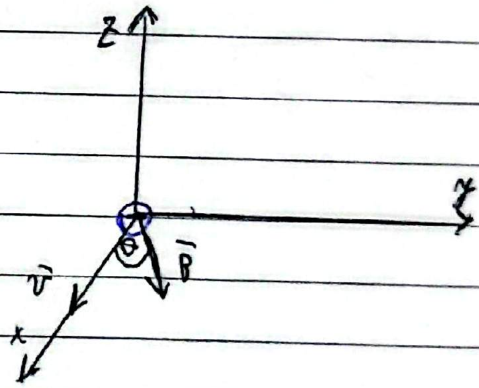
$F = q \cdot (-3 \cdot 10^{5}) = 1,6 \cdot 10^{-19} C \cdot (-3 \cdot 10^{5}) \hat{j} = \boxed{-4,8 \cdot 10^{-14}\ \hat{j}}$

2)-

$\vec{v} = 8 \cdot 10^{6}\ m/s$

$B = 0,025T$

$\theta = 60°$



$|F_B| = |q| \cdot v \cdot B \cdot Sen\theta$

$|F_B| = |1,6 \cdot 10^{-19} c| \cdot (8 \cdot 10^{6}\ m/s)(0,025T) \cdot Sen(60°)$

$|F_B| = 2,77 \cdot 10^{-14}\ \hat{k}$

pero como es un electrón la fuerza va hacia el otro lado

$F_B = \boxed{-2,77 \cdot 10^{-14}\ \hat{k}}$

2)-

$m = 8{,}9 \cdot 10^{-27}$ kg

$q = -6{,}8 \cdot 10^{-19}$ C

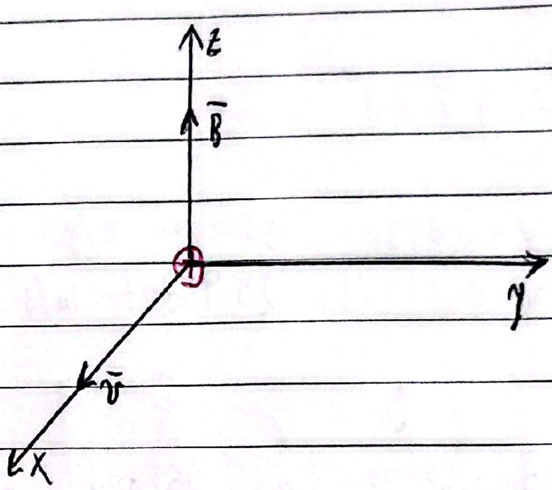
$B = 0{,}7$ T

$v = 7{,}8 \cdot 10^{7}$ m/s

$$r = \frac{m \cdot |v|}{|q||B|} = \frac{8{,}9 \cdot 10^{-27}\,kg \cdot 7{,}8 \cdot 10^{7}\,m/s}{6{,}8 \cdot 10^{-19}\,C \cdot 0{,}7\,T}$$

$\boxed{r = 1{,}46\ m}$

3)-



$q = +1{,}6 \cdot 10^{-19}$ C

$v = 4{,}46 \cdot 10^{6}$ m/s

$B = 1{,}75$ T

$$v \times B = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4{,}46 \cdot 10^{6} & 0 & 0 \\ 0 & 0 & 1{,}75 \end{vmatrix}$$

$v \times B = -7{,}8 \cdot 10^{6}\ \hat{j}$ m·T/s

$F_B = 1{,}6 \cdot 10^{-19}\,C \cdot (-7{,}8 \cdot 10^{6}\ m \cdot T/s)$

$\boxed{F_B = -1{,}248 \cdot 10^{-12}}$

HOJA N°:

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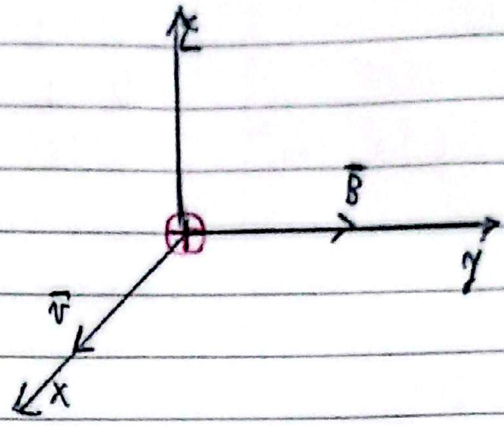
5)-

$q = +1,6 \cdot 10^{-19}\,C$

$v = 12,4\ km/s = 12400\ m/s$

$B = 0,85\,T$



$$\vec{v} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 12400\,m/s & 0 & 0 \\ 0 & 0,85\,T & 0 \end{vmatrix}$$

$\vec{v} \times \vec{B} = 10540\ \hat{k}\ m \cdot T/s$

$F_B = q \cdot 10540\ \hat{k}\ m \cdot T/s$

$F_B = 1,6 \cdot 10^{-19}\,C \cdot 10540\ \hat{k}\ m \cdot T/s$

$\boxed{F_B = 1,6864 \cdot 10^{-15}\ \hat{k}\ N}$

6)-

$q = \cancel{\ }$

$m = 2,58 \cdot 10^{-15}\,kg$

$B = -0,120\,T\ \hat{k}$

$v = 1,05 \cdot 10^{6}\ m/s\,(-3\hat{i} + 4\hat{j} + 12\hat{k})$

$F_B = 2,45\,N$

$$\vec{v} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3,15 \cdot 10^{6} & 4,2 \cdot 10^{6} & 12,6 \cdot 10^{6} \\ 0 & 0 & -0,120 \end{vmatrix}$$

$\vec{v} \times \vec{B} = -504000\ \hat{i} - 378000\ \hat{j}$

$|\vec{v} \times \vec{B}| = \sqrt{(-504000)^2 + (-378000)^2}$

$|\vec{v} \times \vec{B}| = 630000$

$|F_B| = |q| \cdot |\vec{v} \times \vec{B}|$

$|q| = \dfrac{F_B}{|\vec{v} \times \vec{B}|}$

$\boxed{|q| = -3,89 \cdot 10^{-6}\ C}$

$F = m \cdot a$

$a = \dfrac{F}{m}$

$F_B = q\,\vec{v} \times \vec{B}$

$F_B = -3,89 \cdot 10^{-6}\,C\,(-504000\ \hat{i} - 378000\ \hat{j})$

$F_B = 1,96\ \hat{i} + 1,47\ \hat{j}\ N$

b)- $\boxed{\dfrac{F_B}{m} = (7,6 \cdot 10^{14}\ \hat{i} + 5,7 \cdot 10^{14}\ \hat{j})\ m/s^2}$

7)-

$L = 2{,}8\ m$

$I = 5\ A$

$|B| = 0{,}390\ T$

$\theta = 60°, 90°, 120°$

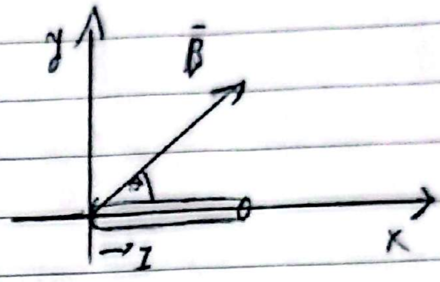
$F_B = I L B \, sen\, \theta$

a- $F_B = 5A \cdot 2{,}8m \cdot 0{,}39T \cdot sen\, 60° = \boxed{4{,}73\ N}$

b- $F_B = 5A \cdot 2{,}8m \cdot 0{,}39T \cdot sen\, 90° = \boxed{5{,}46\ N}$

c- $F_B = 5A \cdot 2{,}8m \cdot 0{,}39T \cdot sen\, 120° = \boxed{4{,}73\ N}$

8)-



$I = 50\ A$

$\theta = 45°$

$B = 1{,}2\ T$

$L = 1\ m$

$B_x = 1{,}2\ T \cdot cos(45°) = 0{,}848$

$B_y = 1{,}2\ T \cdot sen(45°) = 0{,}848$

$F_B = I\, L \times B$

$$L \times B = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 0 \\ 0{,}848 & 0{,}848 & 0 \end{vmatrix}$$

$L \times B = 0{,}848\ \hat{k}$

$F_B = 50A\,(0{,}848\ \hat{k})$

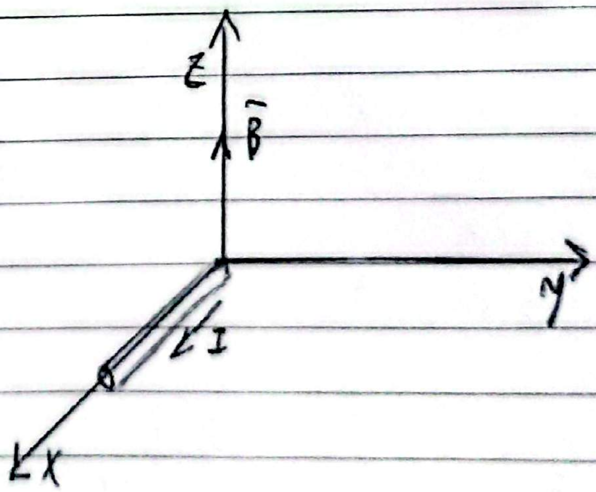
a) $\boxed{F_B = 42{,}4\ \hat{k}\ N}$

b) $|F_B| = I\, L\, B\, sen\, \theta$ p/ $sen\, 90° = máx$

$|F_B| = 50A \cdot 1m \cdot 1{,}2T = \boxed{60\ N}$

SE → la varilla ⊥ a $\bar{B}$.

9)-



$I = 2{,}40\ A$

$L = 0{,}75\ m$

$B = 1{,}60\ \hat{k}\ T$

$$L \times B = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0{,}75m & 0 & 0 \\ 0 & 0 & 1{,}6\ T \end{vmatrix}$$

$L \times B = -1{,}2\ \hat{j}$

$F_B = 2{,}4A \cdot (-1{,}2\ \hat{j}\ m \cdot T)$

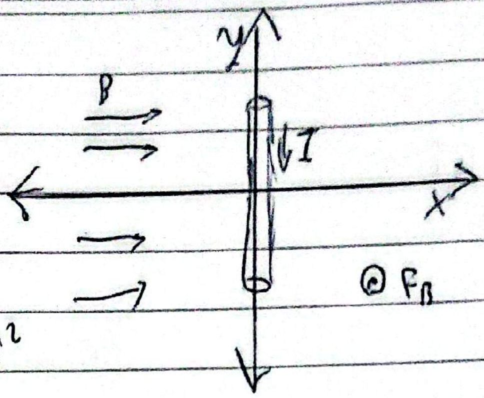
$\boxed{F_B = -2{,}88\ \hat{j}\ N}$

		HOJA Nº:
		FECHA:

1a)-

$m = 0,5\ g/cm = 0,05\ kg/m$

$I = 2A$



$F_B = I \cdot L \cdot |B| \cdot \overset{1}{\overline{\sin(90°)}} = m \cdot |g|$

$$|B| = \frac{m \cdot |g|}{I \cdot L} = \frac{0,05\ kg/m \cdot 9,8\ m/s^2}{2A \cdot 1\ m}$$

$\boxed{|B| = 0,245\ T}$

1b)-

$\tau = N \cdot I \cdot A \cdot B \cdot \sin\theta \rightarrow 1/\theta = 90 \rightarrow \tau_{max}$

$N = 600$

$A = 0,05\ m \times 0,12\ m = 6 \cdot 10^{-3}\ m$

$I = 10^{-3}\ A$

$B = 0,1\ T$

$\tau_{max} = 600 \cdot 6 \cdot 10^{-3}\ m \cdot 10^{-3}\ A \cdot 0,1\ T$

$\boxed{\tau_{max} = 3,6 \cdot 10^{-6}\ N \cdot m}$

2)-

$A = 0,054\ m \cdot 0,085\ m = 4,59 \cdot 10^{-3}$

$N = 25$

$I = 15\ mA = 15 \cdot 10^{-3}\ A$

$B = 0,35\ T$

a- $\mu = N I A$

$\mu = 25 \cdot 15 \cdot 10^{-3}\ A \cdot 4,59 \cdot 10^{-3}\ m^2$

$\boxed{\mu = 1,72 \cdot 10^{-3}\ A \cdot m^2}$

b)-

$|\tau| = \mu \cdot B = (1,72 \cdot 10^{-3}\ A \cdot m^2)(0,35\ T)$

$\boxed{\tau = 6,02 \cdot 10^{-4}\ N \cdot m}$

13)-

$A = 8\text{ cm diámetro} = \pi \cdot 0{,}04^2 = 5{,}03 \cdot 10^{-3}\ m^2$

$N = 12$

$I = 5A$

$B = 6T$

a)- $\tau_{max} = N \cdot I \cdot A \cdot B \cdot \overbrace{\operatorname{sen} 90°}^{1}$

$\tau = 12 \cdot 5A \cdot 5{,}03 \cdot 10^{-3}\ m^2 \cdot 6T$

$\boxed{\tau = 1{,}81\ N \cdot m}$

$\tau_{mitad} = N \cdot I \cdot A \cdot B \cdot \frac{1}{2}$

$\operatorname{sen} \alpha = \frac{1}{2} \Rightarrow \alpha = \operatorname{arcoseno}\left(\frac{1}{2}\right)$

$\boxed{\alpha = 30°}$

$\tau_{mitad} = N \cdot I \cdot A \cdot B \cdot \operatorname{sen}(30°)$

14)-

$r = 0{,}05\ m \Rightarrow 7{,}85 \cdot 10^{-3}\ m^2$

$N = 30$

$I = 5A$

$B = 1{,}2T$

a)- $\mu = N \cdot I \cdot A = 30 \cdot 5A \cdot 7{,}85 \cdot 10^{-3}\ m^2$

$\mu = 1{,}18\ A \cdot m^2$

$\tau = \mu \cdot B = 1{,}18\ A \cdot m^2 \cdot 1{,}2T$

$\boxed{\tau = 1{,}416\ N \cdot m}$